

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

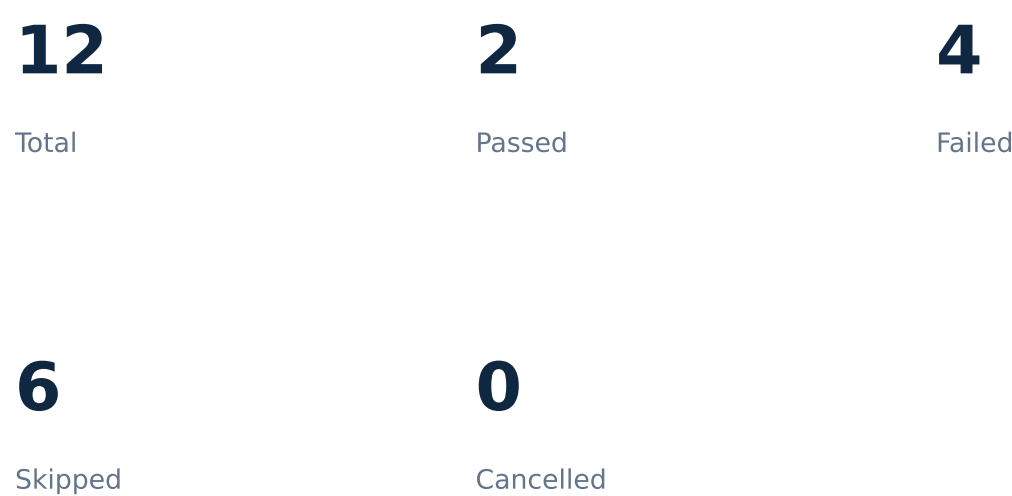
20260813_032934_670b42

Date

2026-08-13T02:29:34.169669+00:00

Executive Summary

20260813_032934_670b42



Overall measured availability: 66.78%

During this campaign, 2 of 12 scenarios passed, 4 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	N/A	N/A	True	N/A	473.492	4010.09	28	3.45	FAIL	N/A	Command timed out after 30 seconds.
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A
cpu	swarm	N/A	N/A	N/A	N/A	665.037	4002.735	2	92.86	FAIL	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	exceeded its scaling EDULABLE: POD_UNSCHE
memory	swarm	N/A	N/A	N/A	Not Required	209.755	1253.894	N/A	100.0	PASS	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	Error response from daemon: Get https://registry-1.docker.io/v2/
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A
container-kill	kubernetes	118.828	135.281	True	True	2010.995	2077.21	44	4.35	PASS	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	lookup registry-1.docker.io on 192.168.65.254:53 Error: dial tcp 192.168.65.254:53: connect: connection refused
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	Error response from daemon: Get https://registry-1.docker.io/v2/
cpu	kubernetes	N/A	N/A	N/A	N/A	18.281	21.621	N/A	100.0	FAIL	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	lookup registry-1.docker.io on 192.168.65.254:53 Error: dial tcp 192.168.65.254:53: connect: connection refused
memory	kubernetes	N/A	N/A	N/A	N/A	16.43	18.103	N/A	100.0	FAIL	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	lookup registry-1.docker.io on 192.168.65.254:53 Error: dial tcp 192.168.65.254:53: connect: connection refused
network-partition	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A
latency	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A
container-kill	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot be restarted on local routes require authentication, but neither PRESERVE nor ESILIENCE_TOKEN is set	N/A

swarm — container-kill

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 28

Availability (%): 3.45

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Command timed out after 30 seconds.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 2

Availability (%): 92.86

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Swarm stress container exceeded its scenario deadline.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

swarm — latency

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — container-kill

State: PASS
Detection Time (s): 118.828

Recovery Time (s): 135.281

HTTP Success: N/A

HTTP Failures: 44

Availability (%): 4.35

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

State: Fail
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

STRESS_POD_UNSCHEDULABLE: ErrImagePull: Error response from daemon: Get "https://registry-1.docker.io/v2/": dial tcp: lookup registry-1.docker.io on 192.168.65.254:53: no such host | Events: Failed to pull image "polinux/stress": Error response from daemon: Get "https://registry-1.docker.io/v2/": dial tcp: lookup registry-1.docker.io on 192.168.65.254:53: no such host | Error: ErrImagePull | Pod sandbox changed, it will be killed and re-created.

Recorded Interpretation

kubernetes — memory

State: **Stale**
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded Interpretation
PRESS POD UNSCHEDULABLE: ErrImagePull: Error response from daemon: Get "https://registry-1.docker.io/v2/": dial tcp: lookup registry-1.docker.io on 192.168.65.254:53: no such host Events: Error: ErrImagePull | Back-off pulling image "polinux/stress" | Error: ImagePullBackOff

kubernetes — network-partition

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

kubernetes — latency

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

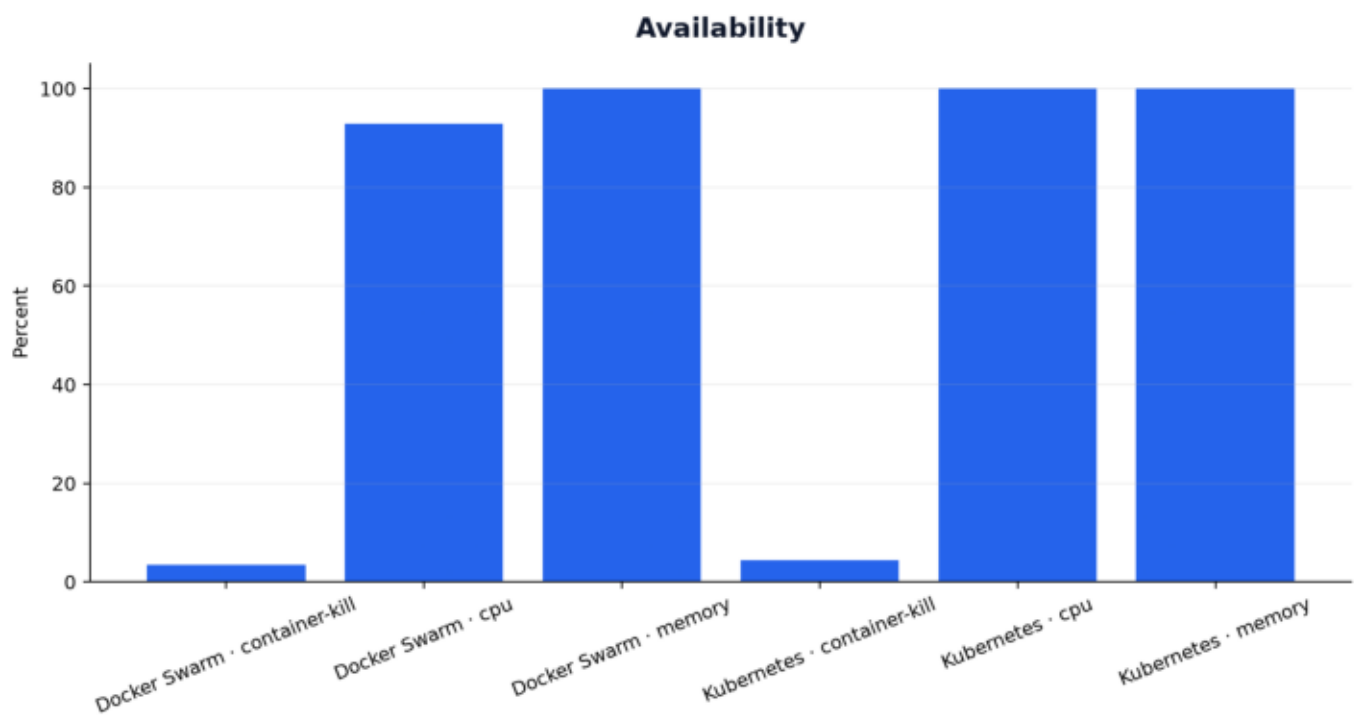
Memory (MB): N/A

Recorded interpretation

Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

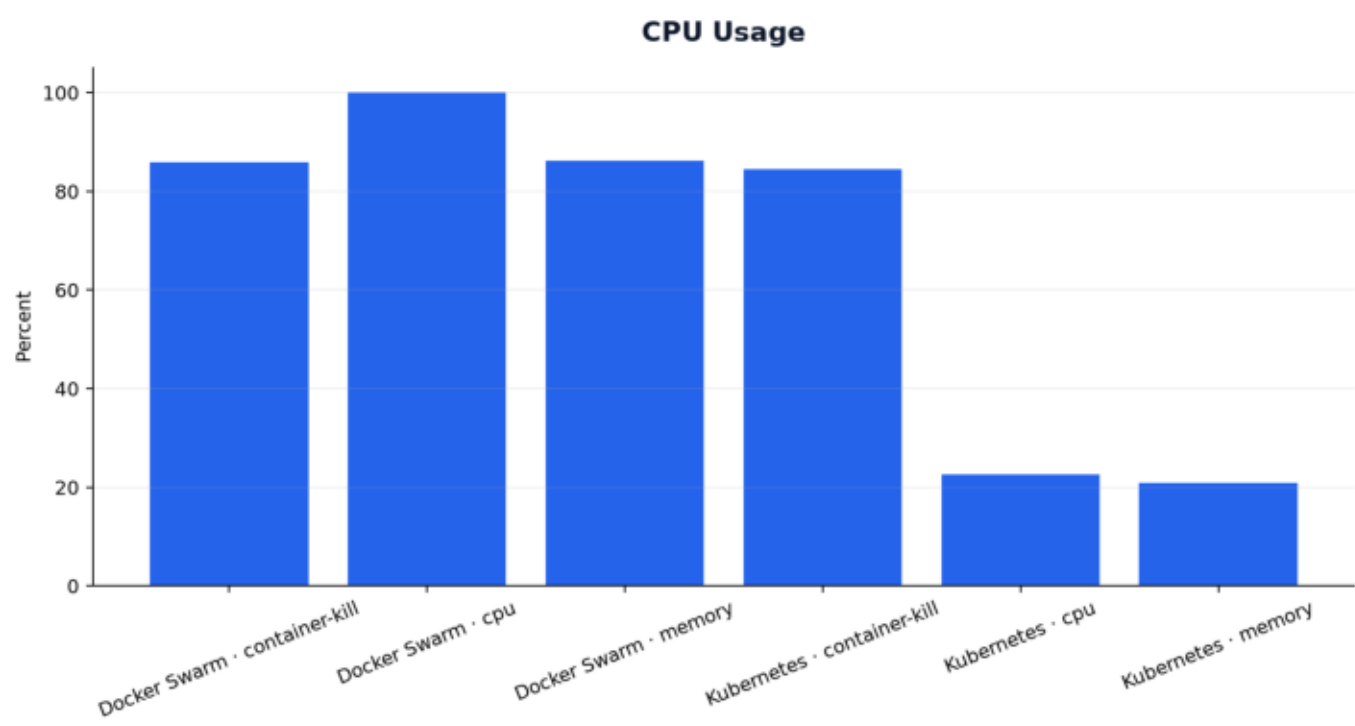
Availability

Current-session measured data



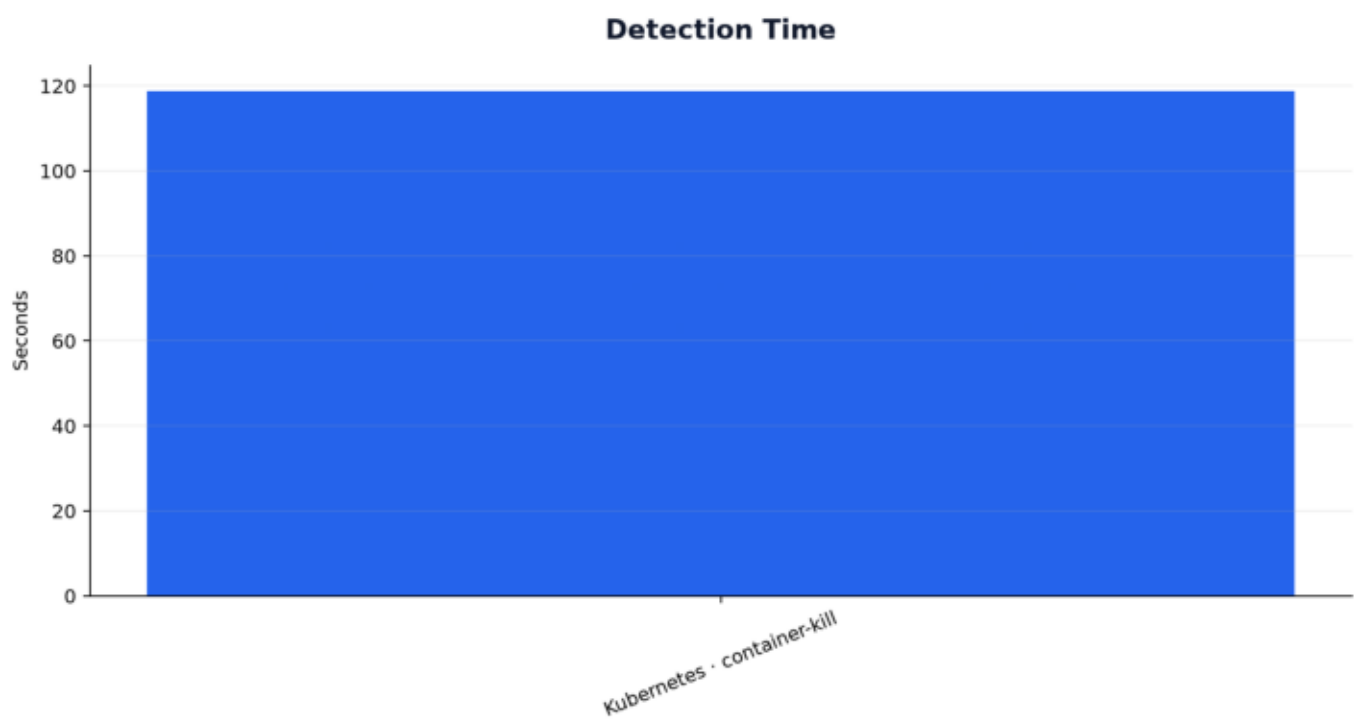
Cpu Usage

Current-session measured data



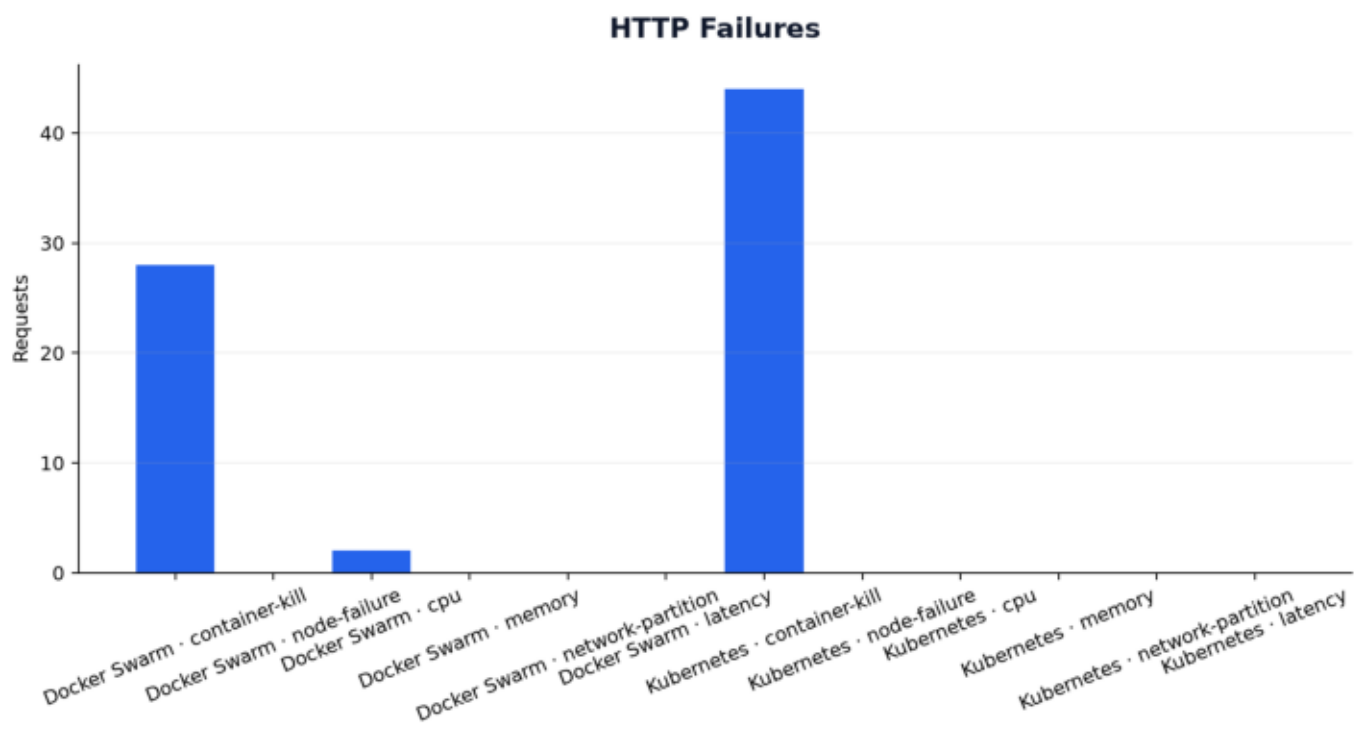
Detection Time

Current-session measured data



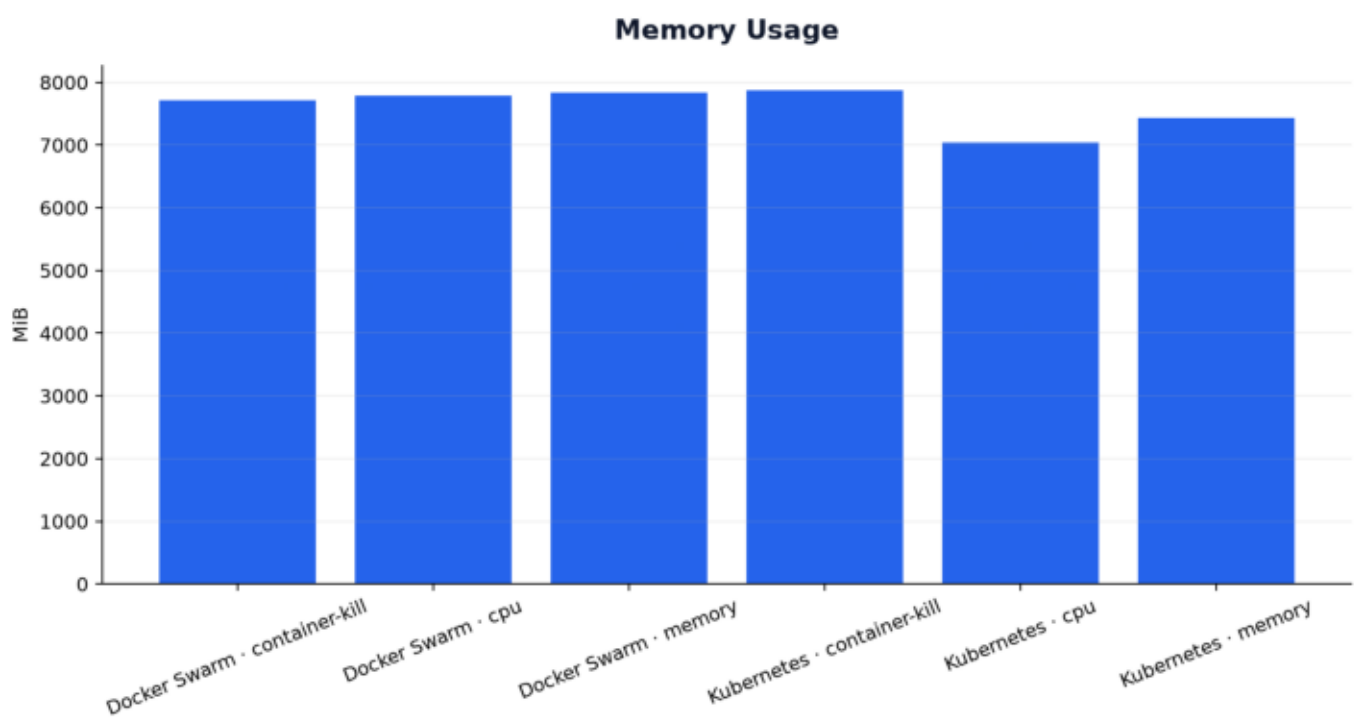
Http Failures

Current-session measured data



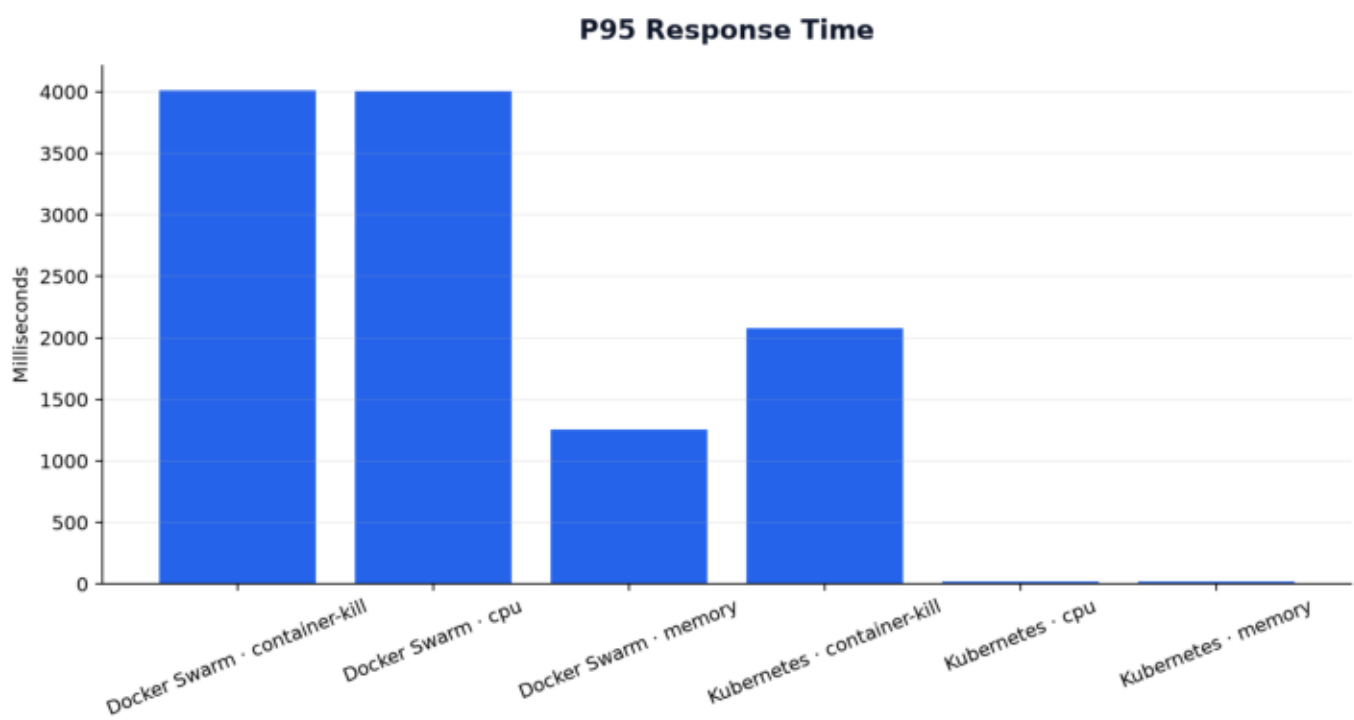
Memory Usage

Current-session measured data



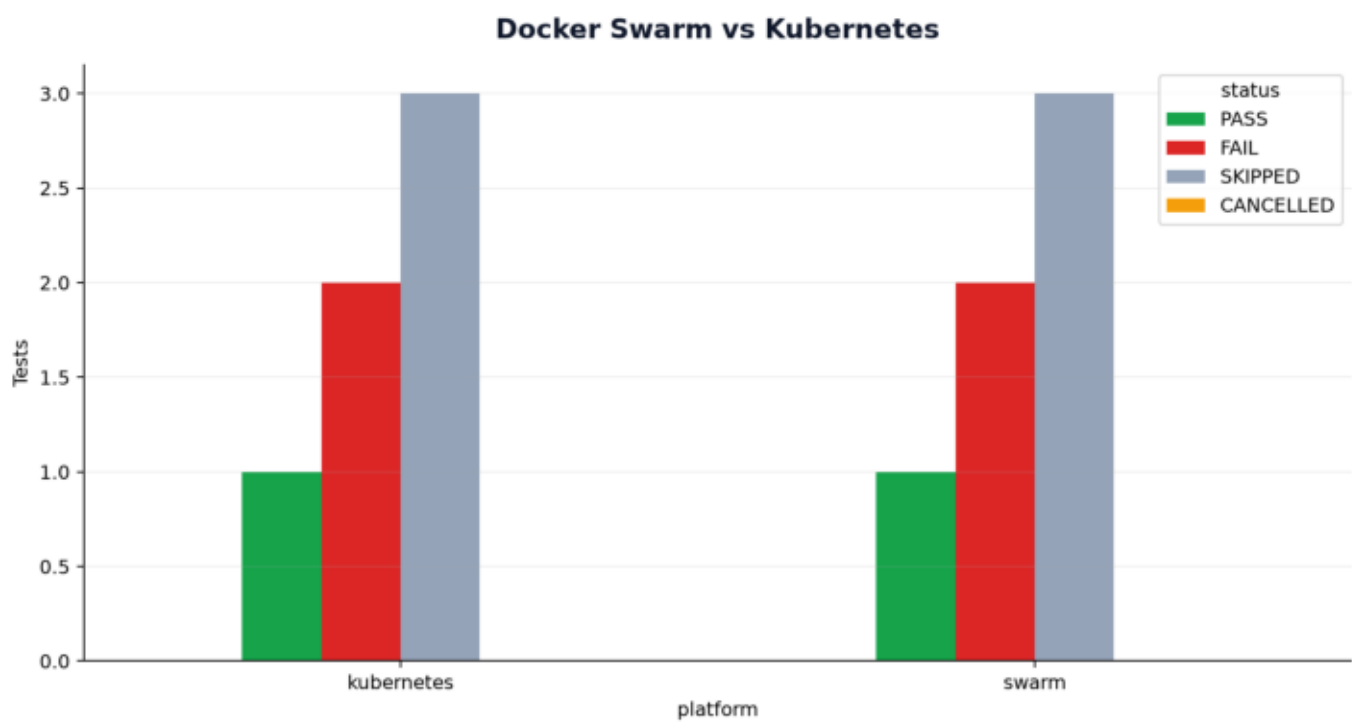
P95 Response Time

Current-session measured data



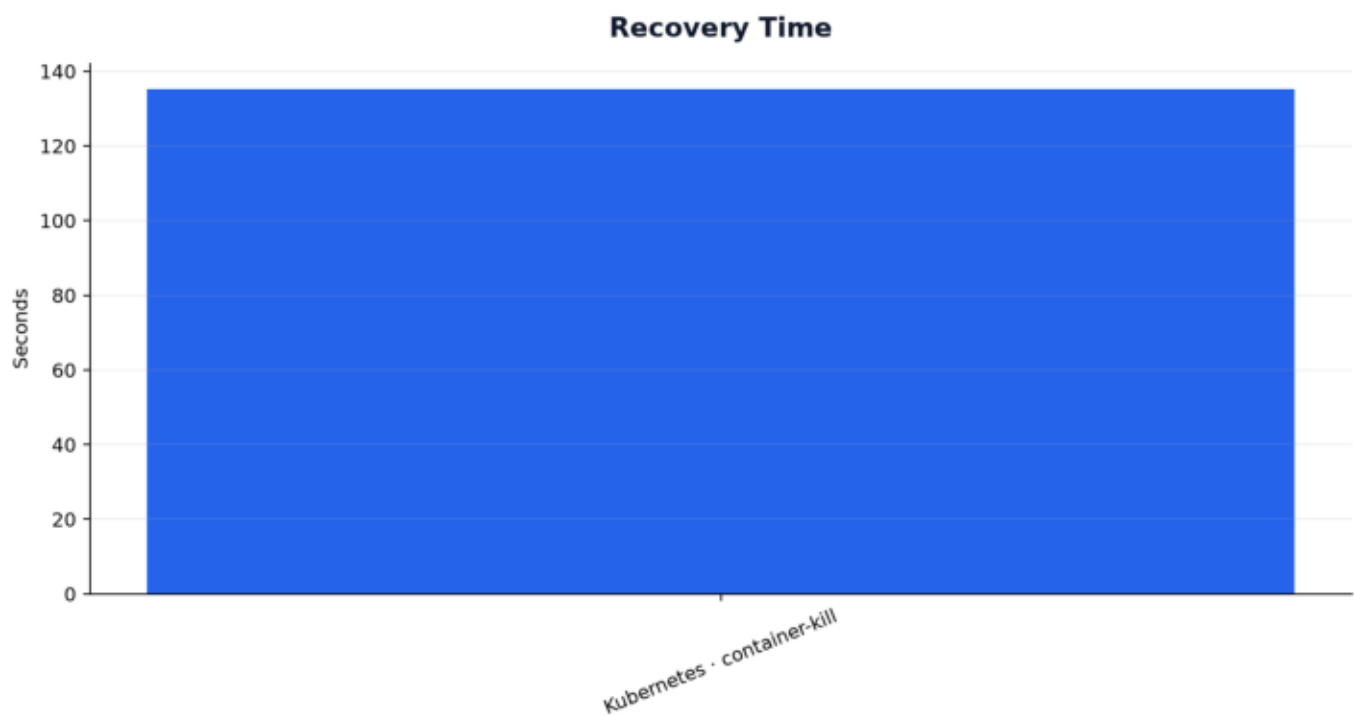
Platform Comparison

Current-session measured data



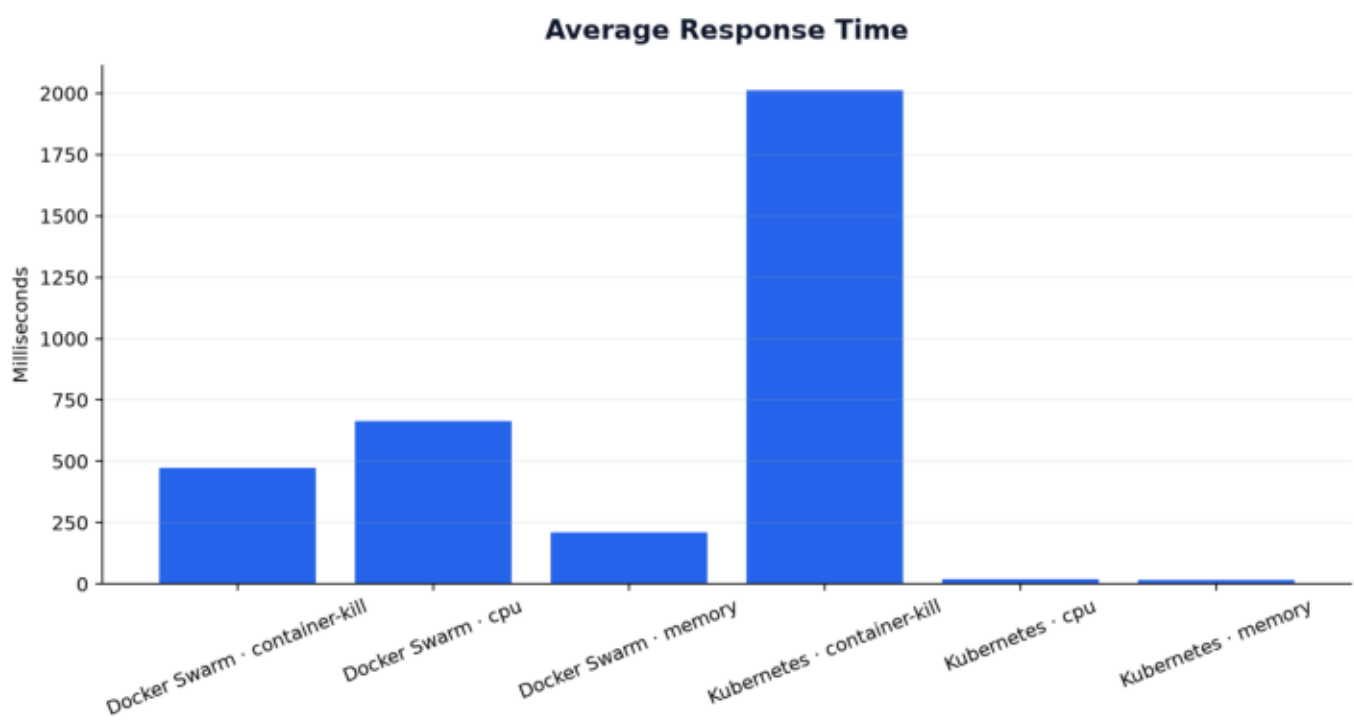
Recovery Time

Current-session measured data



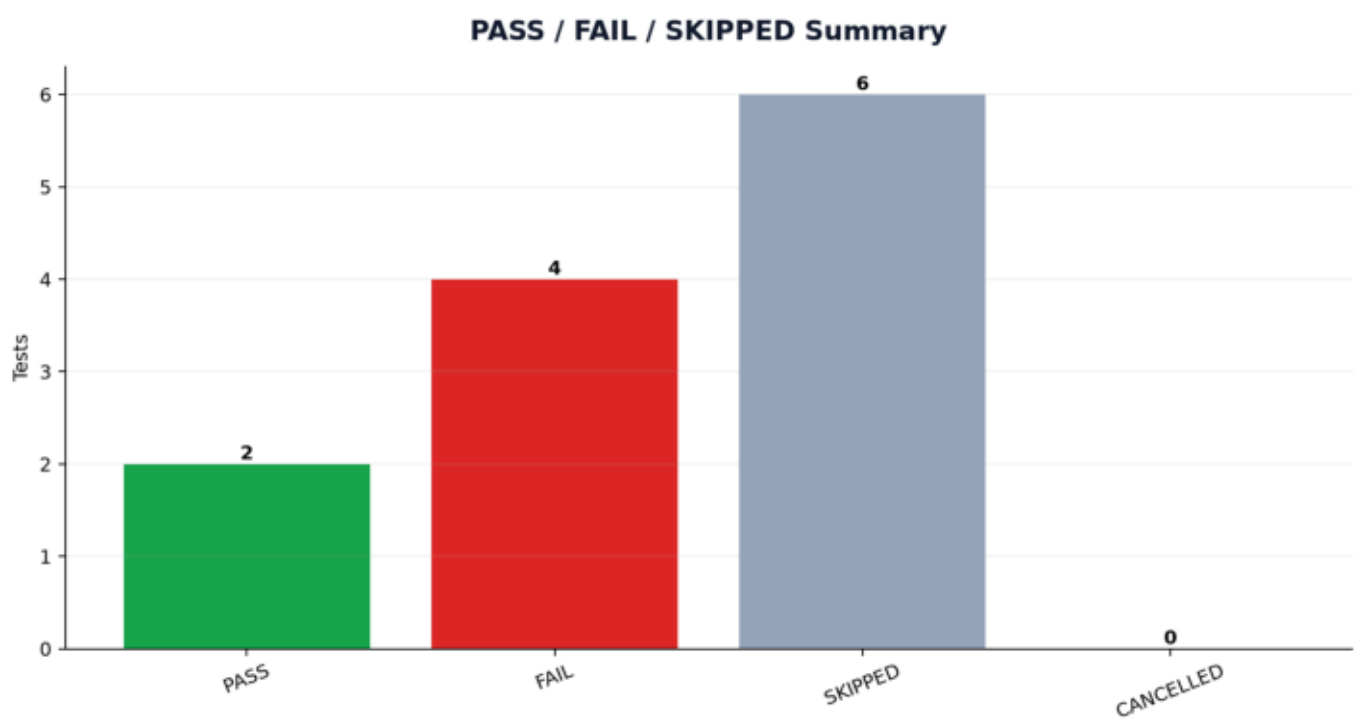
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	container-kill	FAIL	STRESS_POD_UNSCHEDULABLE: Error: Command timed out after 30 seconds: response from daemon: Get https://registry-1.docker.io/v2/: dial tcp 192.168.65.254:53: no such host Error: ImagePullFailed: Pulling back to the daemon
swarm	cpu	FAIL	Swarm stress container exceeded its deadline
kubernetes	cpu	FAIL	STRESS_POD_UNSCHEDULABLE: Error: response from daemon: Get https://registry-1.docker.io/v2/: dial tcp 192.168.65.254:53: no such host Error: ImagePullFailed: Pulling back to the daemon
kubernetes	memory	FAIL	STRESS_POD_UNSCHEDULABLE: Error: response from daemon: Get https://registry-1.docker.io/v2/: dial tcp 192.168.65.254:53: no such host Error: ImagePullFailed: Pulling back to the daemon

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling. Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	network-partition	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
swarm	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	node-failure	Minikube inter-node CN/DNS does not provide Database probe routes. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	network-partition	Database probe routes require authentication, but neither RESILIENCE_API_TOKEN nor RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401
kubernetes	latency	CE_TEST_IDENTIFIER/RESILIENCE_TEST_PASSWORD is configured. Observed: /cars=public:HTTP 401, /renters=public:HTTP 401, /rentals=public:HTTP 401

Swarm vs Kubernetes Comparison

Platform	FAIL	PASS	SKIPPED
kubernetes	2	1	3
swarm	2	1	3

Final Conclusion

20260813_032934_670b42

During this campaign, 2 of 12 scenarios passed, 4 failed and 6 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.